For the navigation model, the existence of a heliport at Kennedy, LaGuardia, Wall Street, Newark, and Morristown was assumed. Further, a precision approach guidance system was simulated at each heliport. This system was modeled after a candidate interim microwave landing system (MLS) which had the operational characteristics summarized in the following table:

Radio transmitter frequency, GHz	15.5
Power (average), W	10
Glide-slope angle (adjustable), deg	2 to 6
Glide-slope beam width (adjustable), deg	1.5 to 2
Localizer coverage, deg	±30
Localizer beam width (adjustable), deg	4 to 6

Orientation and details of the approach and route segments associated with each terminal are given in figure 4. It should be noted that an M prefixed to the standard airport designator was used in figure 4 to designate the simulated instrument landing system; for example, MLGA indicated the simulated approach system at LaGuardia. The geometric characteristics simulated for the guidance system are illustrated in figure 5, along with the coverage window at the Category I decision height of 61 m (200 ft).

A wind model, consisting of a steady-state wind component on which random gusts were superimposed, was developed for this study. The wind velocities selected were between 10 and 15 knots, and the direction was progressively varied in 30° increments prior to the initiation of each circuit to minimize any data bias caused by wind direction. Figure 6, which is a plot of the data obtained from reference 2, shows that the wind velocities seldom exceed the wind spectrum chosen for the simulation. A Dryden gust model (refs. 3 and 4) was used to generate gust disturbances by passing the output of a white-noise generator through three shaping filters, one for each aircraft body axis. The root mean square of the resultant gust intensity was approximately 2.6 knots.

Test Procedure

Six pilots were used for the data runs: two were NYA senior captains, two were NASA research pilots, and two were aeronautical engineers who had operational piloting experience in helicopters. Each pilot received approximately 1 hour of training time on the simulator before starting his data runs.

Each pilot flew the entire route structure, consisting of six segments, twice. The initial conditions consisted of an airspeed of 70 knots at an altitude of 30 m (100 ft) over JFK. The pilot was directed to climb to an altitude of 335 m (1100 ft) while accelerating to 120 knots, navigate by VOR or NDB to the next terminal, reduce airspeed to 70 knots,